Install necessary libraries

! pip install chromadb

! pip install sentence-transformers

```
Collecting chromadb
  Downloading chromadb-1.5.6-cp39-abi3-many1inux_2_17_x86_64.many1inux2Ø14_x86_64.wh1.metadata (5.0 kB)
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Collecting durationpy>=e.7 (from kubernetes>=28.1.e->chromadb)

Downloading durationpy-Ø.10-py3-none-any.wh1. metadata (340 bytes)

```
# Import libraries import chromadb from
sentence_transformers import SentenceTransformer
```

```
client = chromadb. Client() collection = client. create_collection
(name="rag_collection", get_or_create=True) print(f"ChromaDB collection
'{collection. name} ' initialized. ")
```

ChromaDB collection ' rag_collection' initialized.

```
model = SentenceTransformer( ' all -MiniLM- L6-v2 ' )
print("SentenceTransformer model 'all-MiniLM-L6-v2 ' loaded . ") /
usr/10ca1/1ib/python3.12/dist- packages/huggingface_hub/utils/_auth . py :
94 : UserWarning:
The secret does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https: //huggi.ngface.co/settings/tokens), set it You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets. warnings .
  warn(
modules.json: 100% 349/349 36.6kB/s]

config_sentence_transformers.json: 100% 116/116 12.1kB/s]

README.md: 10.5k/? 889kB/s]

sentence_bert_config.json: 100% 53.0/53.0 870B/s]
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING: huggingface_hub.uti1s._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en

```
config.json: 100% 612/612 15.1kB/s]
model.safetensors: 100% 90.9M/90.9M 209MB/s]

Loading weights: 100% 103/103 379.47it/s, Materializing param=pooler.dense.weight]
Bert-model LOAD REPORT from: sentence-transformers/ all-MiniLM- L6-v2
Key I Status
embeddings. position_ids I UNEXPECTED I
Notes :
- UNEXPECTED : can be ignored when loading from different task/ architecture; not ok if you expect identical arch.
tokenizer_config.json: 100% 350/350 10.9kB/s]

vocab. txt: 232k!? 5.16MB/s]

tokenizer.json: 466k/? 15.9MB/s]

special_tokens_map.json: 100% 112/112 1.56kB/s]

config.json: 100% 190/190 4.59kB/s]
Sentence Transformer model ' all-MiniLM- L6-v2 ' loaded .
```

```
documents
    "Artificial Intelligence is the simulation of human intelligence. "
    "Machine Learning is a subset of Al. " ,
    "Deep Learning uses neural networks. " ,
    "Natural Language Processing deals with text data. '
    "RAG combines retrieval and generation . "

ids = ["docl", "doc 2", "doc 3", "doc4" , "doc5"]
print("Samp1e documents prepared.
```

Sample documents prepared.

embeddings model. encode (documents) print(f"Converted {len(documents)} documents into {embeddings. shape[l]}-dimensional embeddings. ")

Converted 5 documents into 384-dimensional embeddings.

collection . add(documents-documents, embeddings=embeddings .tolist() , # Convert numpy array to list for ChromaDB ids=ids

print(f"Added {len(documents)} documents and their embeddings to ChromaDB collection ' {collection. name}' . ")

Added 5 documents and their embeddings to ChromaDB collection ' rag_collection

query = "What is AI?"
# Encode the query into an embedding
query_embedding = model. encode([query])

Perform similarity search in ChromaDB results collection . query(query_embeddings-query_embedding.tolist(), # Convert numpy array to list n_resu1ts=2 # Retrieve the top 2 most similar documents

```
print("Similarity search results:")
print(results)
```

```
Similarity search results:
{'ids': [['doc1', 'doc2']], 'embeddings': None, 'documents': [['Artificial Intelligence is the simulation of human intelligence.
```

```
print("Retrieved Documents:")
for doc in results['documents'][0]:
    print(f"- {doc}")
```

```
Retrieved Documents:
- Artificial Intelligence is the simulation of human intelligence.
- Machine Learning is a subset of AI.
```

```
def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(
        query_embeddings=query_embedding.tolist(),
        n_results=2 # You can adjust this number
    )
    return results['documents'][0]

print("Testing retrieval function for 'Explain machine learning':")
retrieved_ml_docs = retrieve("Explain machine learning")
for doc in retrieved_ml_docs:
    print(f"- {doc}")
```

```
Testing retrieval function for 'Explain machine learning':
- Machine Learning is a subset of AI.
- Deep Learning uses neural networks.
```